

SDLC :- Software Development Life Cycle

It is a procedure to develop the software.

It is a process of creating or altering systems and the models and methodologies that people use to develop these systems.

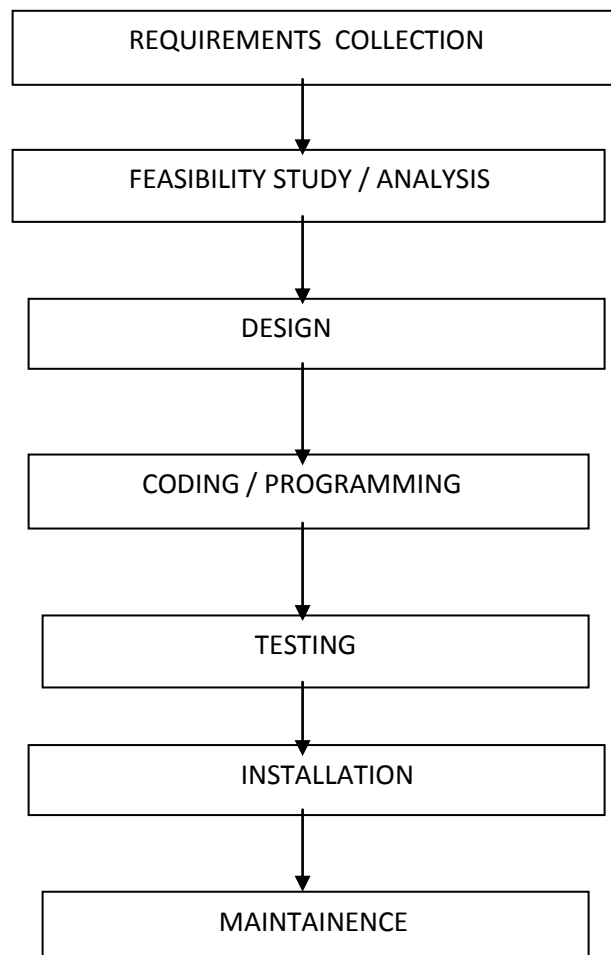
Any SDLC should result in a high quality system that meets or exceeds customer expectations, reaches completion within time and cost estimates, works effectively and efficiently and is inexpensive to maintain and cost effective to enhance.

Different procedures / models are available to develop a software namely,

1) Waterfall model

It is a traditional model

It is a sequential design process, often used in SDLC, in which the progress is seen as flowing steadily downwards (like a waterfall), through the different phases as shown in the figure,



Requirements Collection :-

- done by Business Analysts and Product Analysts
- gathering requirements
- translates business language into software language

For ex, let us consider the example of a banking software.

Feasibility Study :-

- done by software team consisting of project managers, business analysts, architects, finance, HR, developers but not testers
- architect – is the person who tells whether the product can be developed and if yes, then which technology is best suited to develop it.
- here we check for,
 - technical feasibility
 - financial feasibility
 - resource feasibility

Design :-

There are 2 stages in design,

HLD – High Level Design

LLD – Low Level Design

HLD – gives the architecture of the software product to be developed and is done by architects and senior developers

LLD – done by senior developers. It describes how each and every feature in the product should work and how every component should work. Here, only the design will be there and not the code.

For ex, let us consider the example of building a house.

Coding / Programming :-

- done by all developers – seniors, juniors, freshers
- this is the process where we start building the software and start writing the code for the product.

Testing :-

- done by test engineers
- it is the process of checking for all defects and rectifying it.

Installation :-

- done by installation engineers
- to install the product at a client's place for using after the software has been developed and tested.

For ex, consider the example of a software to be developed and installed at Reliance petrol bunk.

Maintenance :-

- here as the customer uses the product, he finds certain bugs and defects and sends the product back for error correction and bug fixing.
- bug fixing takes place
- minor changes like adding, deleting or modifying any small feature in the software product

100 % testing is not possible – because, the way testers test the product is different from the way customers use the product.

Service – based companies and Product – based companies**Service – based companies :-**

They provide service and develop software for other companies

They provide software which is and specified as per the client company's requirement and never keep the code of the developed product and does not provide the software to any other company other than the client company.

Ex – Wipro, Infosys, TCS, Accenture

Product – based companies :-

The develop software products and sell it to many companies which may need the software and make profits for themselves

They are the sole owners of the product they develop and the code used and sell it to other companies which may need the software.

Ex – Oracle, Microsoft

Drawbacks of Waterfall Model :-

In waterfall model, backtracking is not possible i.e, we cannot back and change requirements once the design stage is reached. Thus the requirements are freezed once the design of the software product is started. Change in requirements – leads to change in design – thus bugs enter the design – which leads to change in code which results in more bugs. Thus the requirements are freezed once the design of the product is started.

Advantage of requirements freezing is we get a stable product because there is no change in design and code.

Drawback of requirements freezing – the customer may not be satisfied if the changes he requires is not incorporated in the product. The end result of waterfall model is not a flexible product.

Major drawback of waterfall model – testing is a small phase which is done after coding. Requirement is not tested, design is not tested, if there is a bug in the requirement, it goes on till the end and leads to lot of re-work.

Advantages of waterfall model – requirements do not change nor does design and code, so we get a stable product.

Applications of waterfall model :-

Used in – developing a simple application

- for short term projects

- whenever we are sure that the requirements will not change

For ex, waterfall model can be used in developing a simple calculator as the functions of addition, subtraction etc and the numbers will not change for a long time.

2) SPIRAL MODEL

The spiral model is shown in the figure in the next page.

Ra- requirements analysis of module A. Similarly with Rb, Rc, Rd.

Da – design of module A. Similarly with Db, Dc, Dd

Ca – coding of module A. Similarly with Cb, Cc, Cd

Ta – testing of module A. Similarly with Tb, Tc, Td

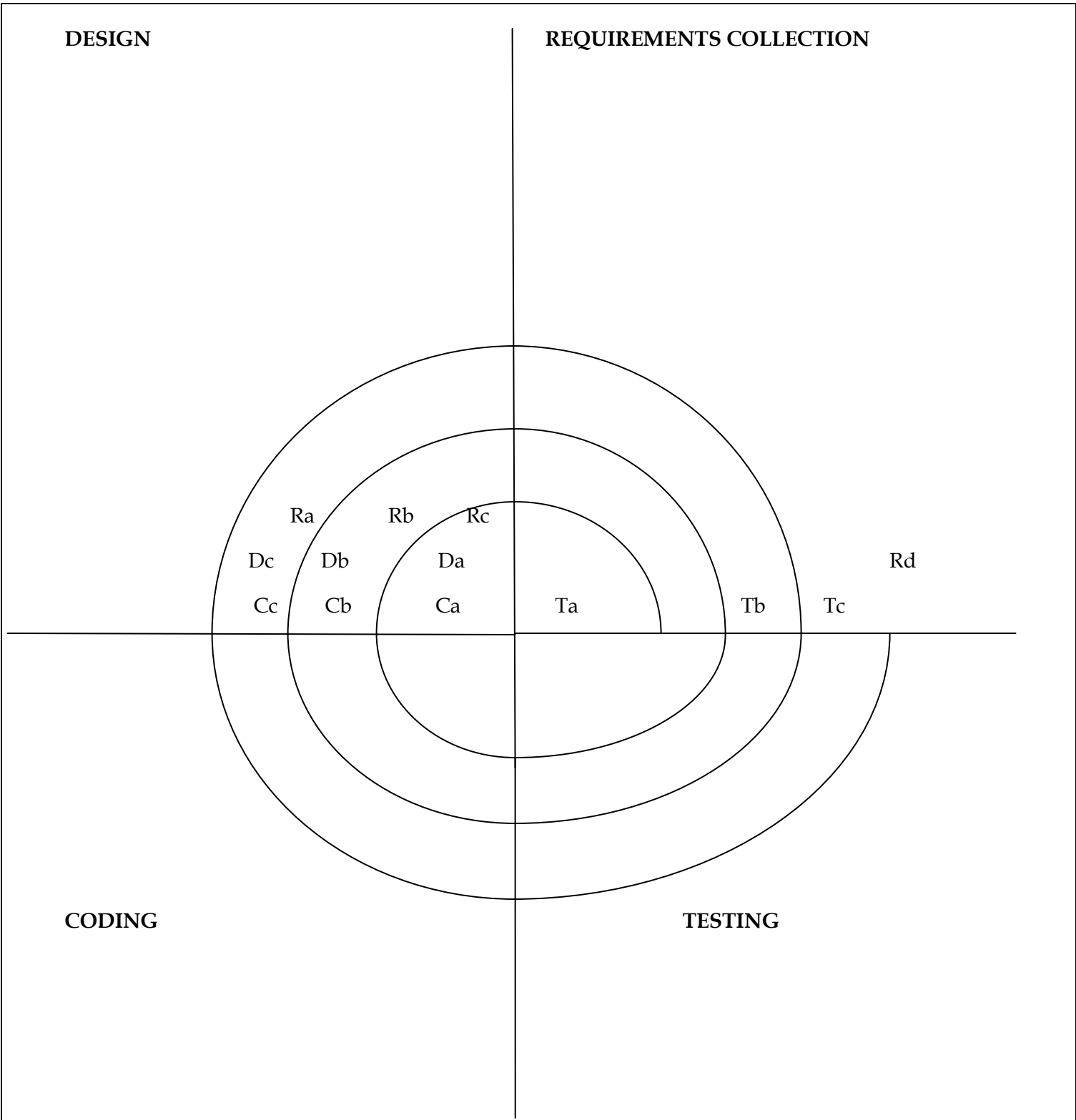
In Spiral model, the software product is developed in small modules. Let us consider the figure shown below in developing a s/w product X. X is built by integrating A,B,C and D.

The module A – requirements of the module is collected first and then the module is designed. The coding of module A is done after which it is tested for defects and bugs.

The module B – once module A has been built, we start the same process for module B. But while testing module B, we test for 3 conditions – a)test module B b)test integration of module B with A c)test module A.

The module C – after building module A,B, we start the same process for module C. Here we test for the following conditions – 1) test module c, b, a 2) test for integration of C and B, C and A, A and B.

And thus the cycle continues for different modules. Thus in the above example, module B can be built only after module A has been built correctly and similarly for module C.



For spiral model, the best example that we can consider is the MS-Excel application.

The MS-Excel sheet consists of a number of cells that are the components of Excel sheet.

Here we have to create the cells first (module A). Then we can do operations on the cells like merge cells into two, split cell into half (module B). Then we can draw graphs on the excel sheet (module C).

Advantages of Spiral Model :-

- 1) Requirement changes are allowed.
- 2) After we develop one feature / module of the product, then and only then we can go on to develop the next module of the product.

Whenever the customer request for major changes in requirements in a particular module, then we change only that module and do testing of both unit and integration of units. This change in requirements comes up in a separate cycle just to do the changes.

Whenever the customer request minor changes in the product, then the s/w team makes the minor changes along with the new module to be developed simultaneously in a single cycle. We don't consider making the minor change in a separate cycle of the spiral model due to time and resource constraints.

The documents collected by Business analysts during requirement collection stage is known as **CRS (Customer Requirement Specification)** or **BRS (Business Requirement Specification)** or **BS (Business Specification)**. In this document, the client explains how their business works or the requirement of the s/w he needs. The BA gathers CRS from the client and translates it into **SRS (Software Requirement Specification)**. The SRS contains how the software should be developed and is given by the BA to developers. For more detailed explanation of how to go about developing the s/w, the BA/developer builds another document - **FS (Functional Specification)**. FS explains how each and every component should work.

Drawbacks of Spiral Model - Traditional model and thus developers only did testing job as well.

Applications of Spiral Model

- whenever there is dependency in building the different modules of the software, then we use Spiral Model.
- whenever the customer gives the requirements in stages, we develop the product in stages.

3) V - MODEL / V & V MODEL (Verification and Validation Model)

This model came up in order to overcome the drawback of waterfall model - here testing starts from the requirement stage itself.

The V & V model is shown in the figure in the next page.

1) In the first stage, the client send the CRS both to developers and testers. The developers translate the CRS to the SRS.

The testers do the following tests on CRS,

1. Review CRS

- a. conflicts in the requirements
 - b. missing requirements
 - c. wrong requirements
2. Write Acceptance Test plan
3. Write Acceptance Test cases

The testing team reviews the CRS and identifies mistakes and defects and send it to the development team for correcting the bugs. The development updates the CRS and continues developing SRS simultaneously.

2) In the next stage, the SRS is sent to the testing team for review and the developers start building the HLD of the product. The testers do the following tests on SRS,

1. Review SRS against CRS
 - a. every CRS is converted to SRS
 - b. CRS not converted properly to SRS
2. Write System Test plan
3. Write System Test case

The testing team reviews every detail of the SRS if the CRS has been converted properly to SRS.

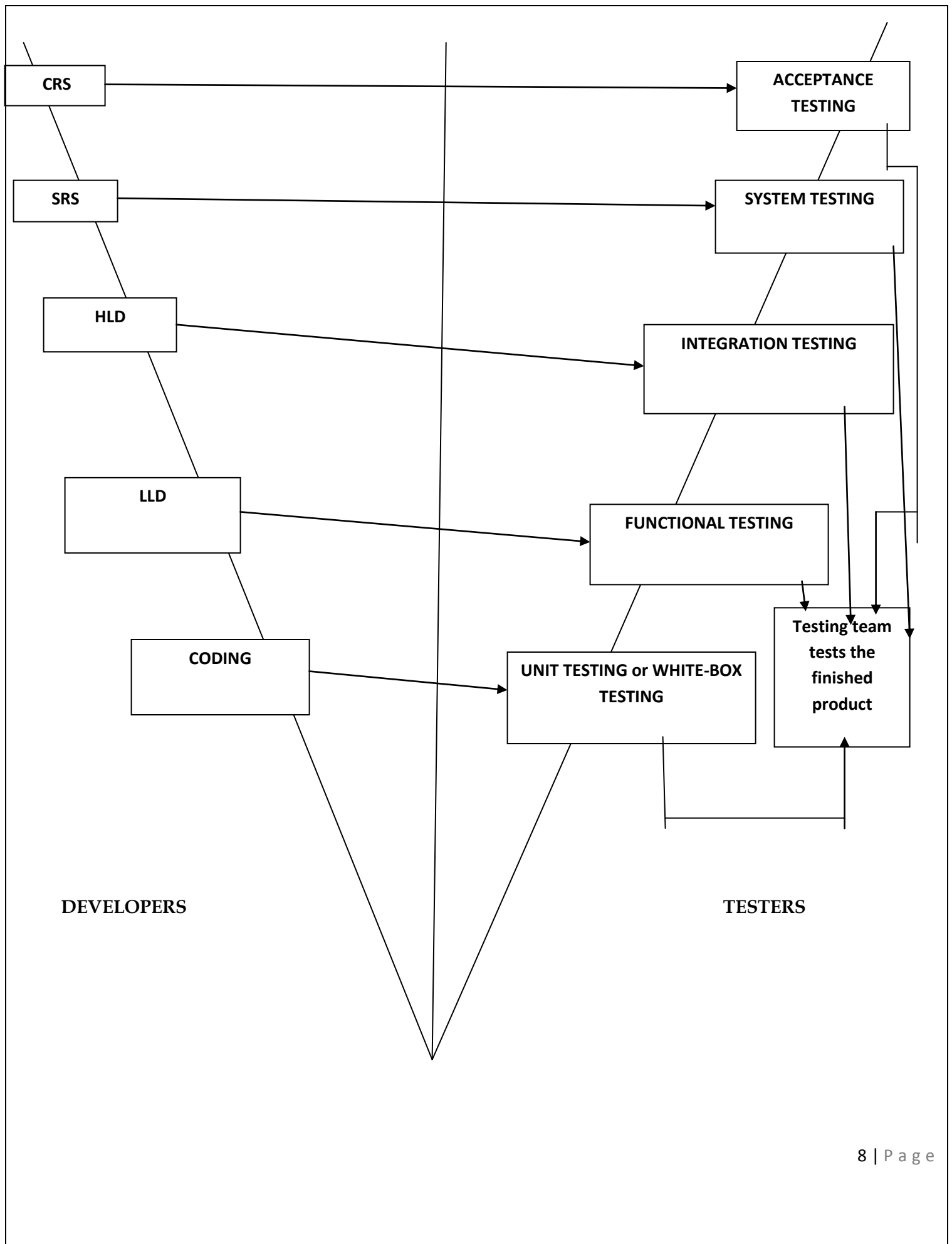
3) In the next stage, the developers start building the LLD of the product. The testers do the following tests on HLD,

1. Review HLD
2. Write Integration test plan
3. Write Integration test case

4) In the next stage, the developers start with the coding of the product. The testing team carries out the following tasks,

1. Review LLD
2. Write Functional test plan
3. Write Functional Test case

After coding, the developers themselves carry out **unit testing** or **also known as white box testing**. Here the developers check each and every line of code and if the code is correct. After white-box testing, the s/w product is sent to the testing team which tests the s/w product and carries out functional testing, integration testing, system testing and acceptance testing and finally deliver the product to the client.



How to handle requirement changes in V&V :-

Whenever there is change in requirement, the same procedure continues and the documents will be updated.

Advantages of V&V model

- 1) Testing starts in very early stages of product development which avoids downward flow of defects which in turn reduces lot of rework
- 2) Testing is involved in every stage of product development
- 3) Deliverables are parallel/simultaneous – as developers are building SRS, testers are testing CRS and also writing ATP and ATC and so on. Thus as the developers give the finished product to testing team, the testing team is ready with all the test plans and test cases and thus the project is completed fast.
- 4) Total investment is less – as there is no downward flow of defects. Thus there is less or no re-work

Drawbacks of V&V model

- 1) Initial investment is more – because right from the beginning testing team is needed
- 2) More documentation work – because of the test plans and test cases and all other documents

Applications of V&V model

We go for V&V model in the following cases,

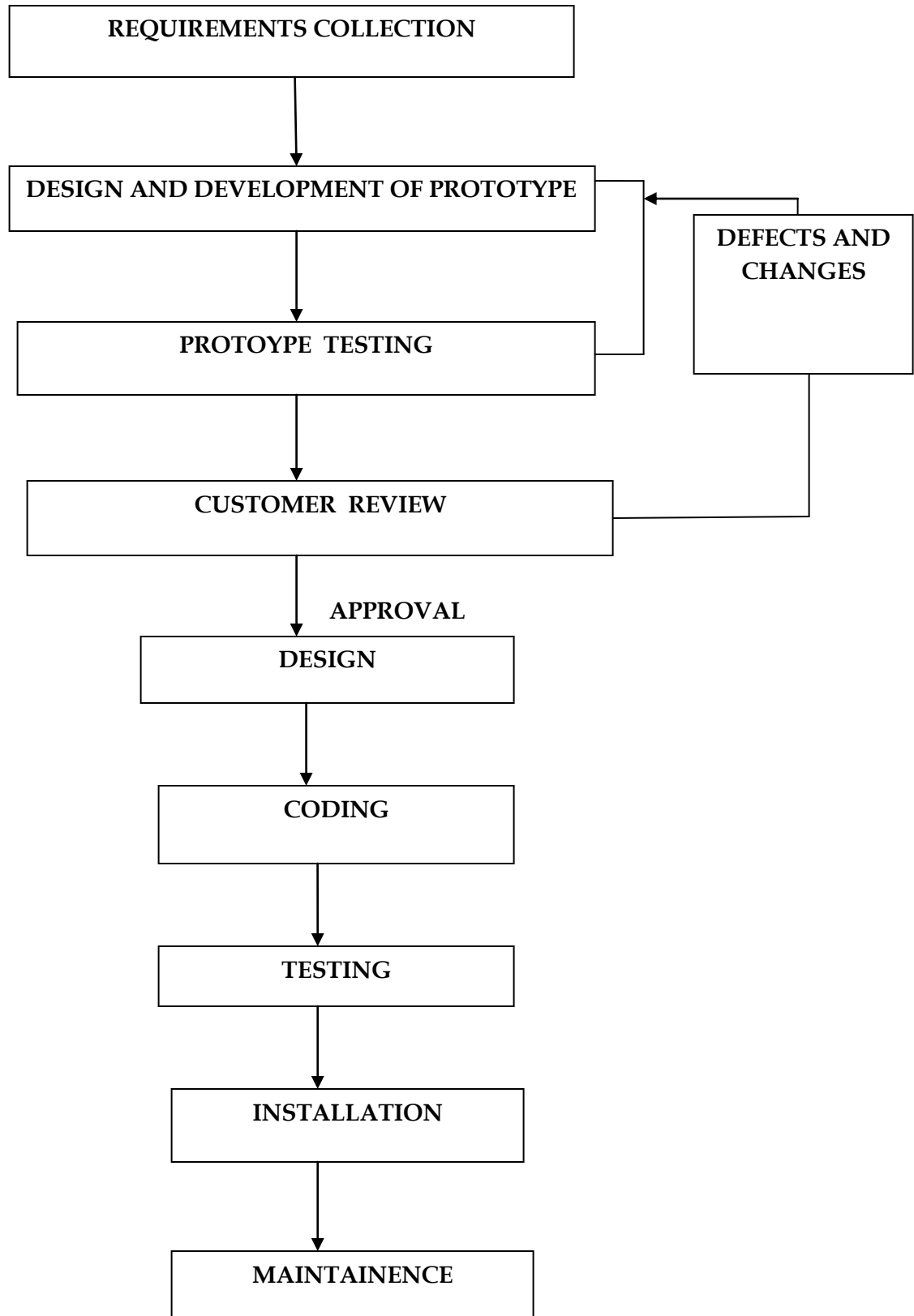
- 1) for long term projects
- 2) for complex applications
- 3) when customer is expecting a very high quality product within stipulated time frame because every stage is tested and developers & testing team are working in parallel

4) PROTOTYPE DEVELOPMENT MODEL

The requirements are collected from the client in a textual format. The prototype of the s/w product is developed. The prototype is just an image / picture of the required s/w product. The customer can look at the prototype and if he is not satisfied, then he can request more changes in the requirements.

Prototype testing means developers/ testers are checking if all the components mentioned are existing.

The difference b/w prototype testing and actual testing – in PTT, we are checking if all the components are existing, whereas, in ATT we check if all components are working.



From “REQUIREMENT COLLECTION” to “CUSTOMER REVIEW”, textual format has been converted to image format. It is simply extended requirement collection stage. Actual design starts from “DESIGN” stage.

Prototype development was earlier done by developers. But, now it is done by web designers/content developers. They develop prototype of the product using simple ready-made tools. Prototype is simply an image of the actual product to be developed.

Advantages of Prototype model

- 1) In the beginning itself, we set the expectation of the client.
- 2) There is clear communication b/w development team and client as to the requirements and the final outcome of the project
- 3) Major advantage is – customer gets the opportunity in the beginning itself to ask for changes in requirements as it is easy to do requirement changes in prototype rather than real applications. Thus costs are less and expectations are met.

Drawbacks of Prototype model

- 1) There is delay in starting the real project
- 2) To improve the communication, there is an investment needed in building the prototype.

Applications

We use this model when,

- 1) Customer is new to the s/w
- 2) When developers are new to the domain
- 3) When customer is not clear about his own requirement

There are 2 types of prototype,

Static Prototype – entire prototype of the requirement is stored in a word document with explanation and snapshots and instructions on how to go about building the s/w, how the finished product will look like and its working etc.

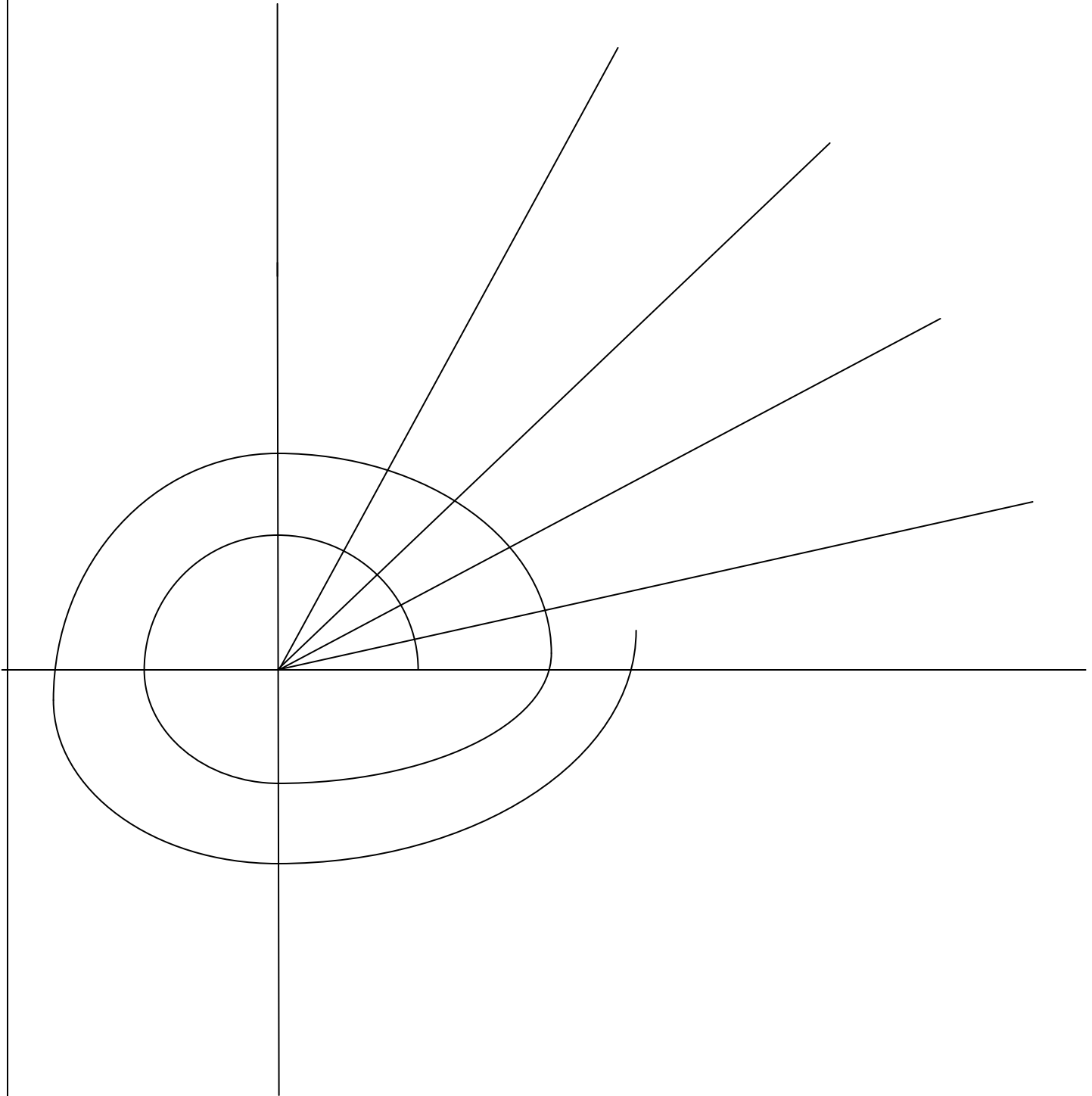
Dynamic Prototype – similar to a browser, but we can't enter any information. Only the features are available without entering data. It's like a dummy page, made out of HTML with tags and links to different pages representing features of the project

5) **Derived model or Customized model** – we can take any of the above 4 models and change it as per business needs and requirements

6) HYBRID MODEL

It combines 2 or more models and modify them as per business requirements.

A) Hybrid model of Spiral and Prototype development models



We go for this model when,

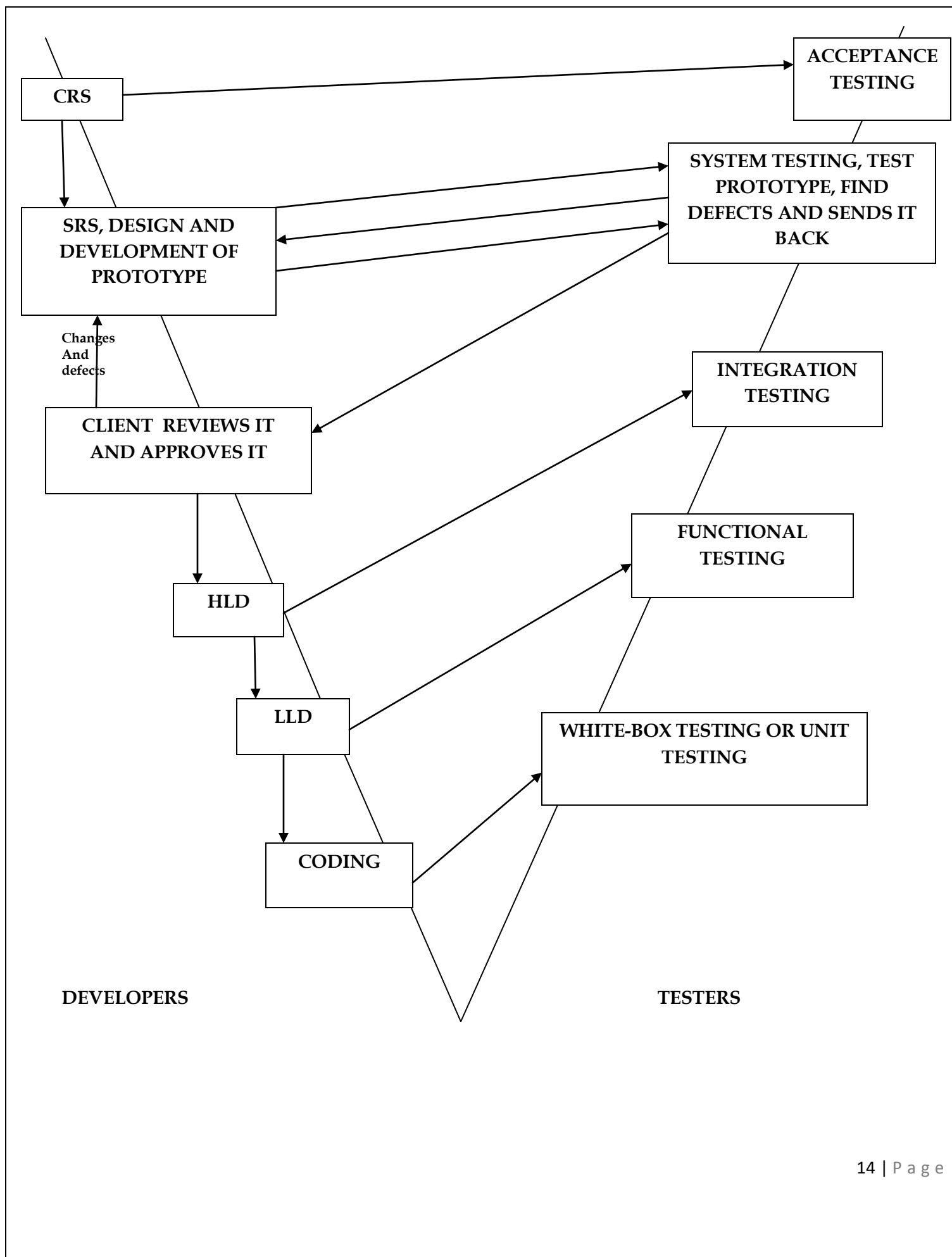
- 1) Whenever there is dependency, we go for this hybrid model
- 2) When the customer gives requirement in stages, we develop the product in stages using this hybrid model.
- 3) When the customer is new to the s/w domain
- 4) When developers are new to the domain
- 5) When customer is not clear about his own requirements

Hybrid model of V&V and Prototype model

We go for this model when,

- 1) Testing starts from early stages of product development which avoids downward flow of defects, thus reducing re-work.
- 2) When customer is expecting a very high quality product within stipulated time frame because every stage is tested and developers and testing team work in parallel.
- 3) When client and developers are both new to the domain
- 4) When customer is not clear about his own requirements

In this hybrid model, the testing team is involved in testing the prototype.



INTERVIEW QUESTIONS

1) What is SDLC ?

2) What are the different models available ?

*ANS) Tell the 1st 6 models and 7) RUP – Rational Unified Process Model 8) Agile Model
9) RAD – Rapid Application Development*

3) Advantages, Disadvantages and Applications of each model

4) A model for every project in our resume -> for 3 projects we do, be prepared to tell which model we used for each project and why we used that particular model only. The most common answer we can tell – we used a hybrid of “so and so” model for the reasons such as – client was not sure of his requirements, etc .